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Regenerative tree growth processes*

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Abstract

We introduce regenerative tree growth processes as consistent families of random trees with n labelled leaves, $n \geq 1$, with a regenerative property at branch points. We establish necessary and sufficient conditions on the growth rules under which we can apply results by Haas and Miermont to establish self-similar random trees and residual mass processes as scaling limits. This framework includes all growth processes for exchangeably labelled Markov branching trees, as well as non-exchangeable models such as the alpha-theta model, the alpha-gamma model and all restricted exchangeable models previously studied. A key result is a representation of the growth rules with a σ-finite dislocation measure extending Bertoin's notion of an exchangeable dislocation measure from the setting of homogeneous fragmentations.

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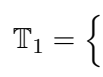
1 Introduction to regenerative tree growth processes

For each $n \geq 1$, denote by $\mathbb{T}_n$ the set of rooted leaf-labelled combinatorial trees with no degree-2 vertices and $n+1$ degree-1 vertices, one of which is called the *root*, the others *leaves*. We distinguish the leaves by labels $1, \ldots, n$. Vertices of degree 3 or higher are called *branch points*. Consider a family T_n, $n \geq 1$, of random trees in $\mathbb{T}_n$, $n \geq 1$. For $n \geq 2$, we refer to the vertex adjacent to the root as the *first branch point*. It induces the *first split*, a random partition $\Pi_n = (\Pi_{n,1}, \ldots, \Pi_{n,K_n})$ of the label set $[n] := \{1, \ldots, n\}$ into the label sets of the *subtrees above the branch point*, the connected components of the tree with the first branch point removed. Here, we put the *blocks* $\Pi_{n,i}$ of Π_n in the order of their least elements. For illustration, we write

$\mathbb{T}_1 =$  , $\mathbb{T}_2 =$, $\mathbb{T}_3 =$, etc.,

where we have ordered subtrees by their least labels to uniquely choose plane tree representatives.

We suppose that the family $(T_n, n \geq 1)$ is *consistent* in the sense that removal of leaf $n+1$ (and the resulting degree-2 vertex, if any) from T_{n+1} yields T_n. Reversing this removal gives a tree growth step from n to $n+1$. A consistent family $(T_n, n \geq 1)$ is called a *tree growth process*. For $B \subseteq [n]$, let $\mathbb{T}_B$ be the set of trees with $\#B$ leaves labelled by B, so that $\mathbb{T}_{[n]} = \mathbb{T}_n$. Let $T_{n,B} \in \mathbb{T}_B$ be the *reduced subtree* of T_n spanned by the root of T_n and leaves in B, and let $\widetilde{T}_{n,B} \in \mathbb{T}_{[\#B]}$ be the image of $T_{n,B}$ after *relabelling of leaves* by the increasing bijection from B to $[\#B]$.

Definition 1 We call a tree growth process $(T_n, n \geq 1)$ *regenerative* if for each $n \geq 2$, conditionally given that the first split of T_n is $\Pi_n = (B_1, \ldots, B_k)$, the subtrees $\widetilde{T}_{n,B_i}$, $1 \leq i \leq k$, are independent copies of $T_{\#B_i}$.

In the terminology of [5], the trees in a regenerative tree growth process as defined here are "consistent labelled Markov branching trees". The *exchangeable* case, where the distribution of T_n

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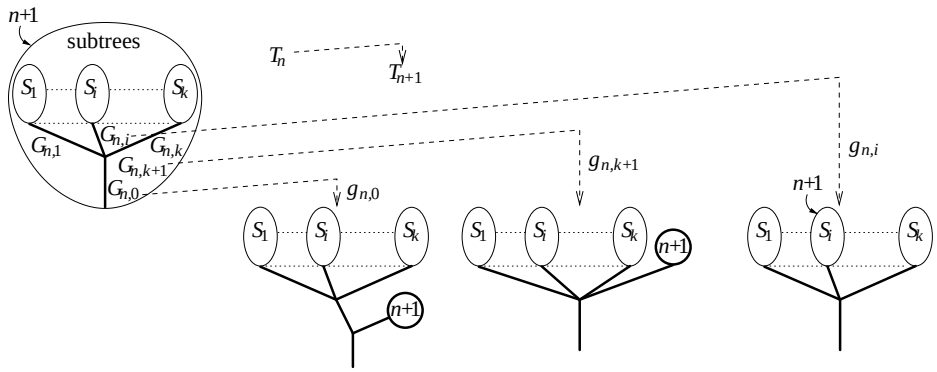

Figure 1: Illustration of the regenerative tree growth step

is invariant under all permutations of labels, was initiated by Aldous [2], who posed the problem of providing a Kingman-type representation in this case. Bertoin's [3] theory of homogeneous fragmentations solved that problem as explained in [15]. Then [15, 16] studied regenerative tree growth processes associated with fragmentation processes. Natural non-exchangeable tree growth processes were described in terms of simple growth rules that admit regenerative descriptions based on the first split and its subtrees, see particularly [4, 10, 23], as reviewed in Examples 3 and 4 below. Such regenerative growth rules are a key consequence of Definition 1 and lead to the following characterisation (cf. Figure 1). We leave the straightforward proof to the reader.

Proposition 2 *In the tree growth step from n to $n+1$ for $n \geq 2$, there are the following disjoint events, $G_{n,i}$ for $i = 0, \ldots, K_n + 1$, where $K_n \geq 2$ is the number of blocks of the first split of T_n:*

- *$G_{n,0}$: leaf $n+1$ is attached to a new branch point between the root and the first branch point of T_n;*
- *$G_{n,i}$, $1 \leq i \leq K_n$: label $n+1$ is inserted into the ith block of the first split;*
- *G_{n,K_n+1}: leaf $n+1$ is attached to the first branch point, as singleton block of the first split.*

A tree growth process $(T_n, n \geq 1)$ is regenerative if and only if $\mathbb{P}(G_{n,0} \,|\, T_n) = \mathbb{P}(G_{n,0})$ does not depend on T_n and $\mathbb{P}(G_{n,i} \,|\, T_n) = \mathbb{P}(G_{n,i} \,|\, \Pi_n)$, $1 \leq i \leq K_n + 1$, only depends on the partition Π_n of the first split. In the event $G_{n,i}$, $1 \leq i \leq K_n$, label $n+1$ is inserted into the ith subtree of T_n of size $\#\Pi_{n,i}$ following the same rules, up to relabelling by the increasing bijection from $\Pi_{n,i} \cup \{n+1\}$ to $[\#\Pi_{n,i} + 1]$.

We denote by $\mathcal{P}_n$ the set of partitions $\pi = (B_1, \ldots, B_k)$ of $[n]$, with blocks B_i ordered by least element. We use notation $g_n(\pi, i) = \mathbb{P}(G_{n,i} \,|\, \Pi_n = \pi)$, $0 \leq i \leq k+1$, for $\pi \neq 1_{[n]} := ([n])$, $n \geq 2$, and write $g_n(0) = g_n(\pi, 0)$, since we require that this quantity is independent of $\pi \in \mathcal{P}_n \setminus \{1_{[n]}\}$.

Example 3 (Alpha-theta model [23]) For $0 \leq \alpha \leq 1$, $\theta \geq 0$ and $\pi = (B_1, B_2) \in \mathcal{P}_n$, let

$$g_n(\pi, 0) = \frac{\alpha}{n-1+\theta}, \; g_n(\pi, 1) = \frac{\#B_1 - 1 + \theta}{n-1+\theta}, \; g_n(\pi, 2) = \frac{\#B_2 - \alpha}{n-1+\theta}, \; g_n(\pi, k) = 0, k \geq 3.$$

Example 4 (Alpha-gamma model [4]) For $0 \leq \gamma \leq \alpha \leq 1$ and $\pi = (B_1, \ldots, B_k) \in \mathcal{P}_n$, let

$$g_n(\pi, 0) = \frac{\gamma}{n-\alpha}, \; g_n(\pi, i) = \frac{\#B_i - \alpha}{n-\alpha}, i \in [k], \; g_n(\pi, k+1) = \frac{(k-1)\alpha - \gamma}{n-\alpha}.$$

In these examples, the regenerative property was shown in [23, Proposition 11] and [5, Proposition 8], respectively. They both contain as special case for $\alpha = 1/2$ and, respectively, $\theta = 1/2$ and $\gamma = 1/2$, the exchangeable uniform model on binary trees, related to Aldous's Brownian Continuum Random Tree [1]. The uniform model on $\mathbb{T}_n$ is not consistent, see [22] for weak limits. Ford's binary alpha-model [10] is also included in both examples.

Let us express the splitting rules $p_n(\pi) = \mathbb{P}(\Pi_n = \pi)$ explicitly in terms of $(g_n, n \geq 2)$. From the growth rule, we have for all $\pi = (B_1, \ldots, B_k) \in \mathcal{P}_n \setminus \{1_{[n]}\}$,

$$\begin{aligned} p_2(\{1\},\{2\}) = 1, \qquad p_{n+1}([n],\{n+1\}) = g_n(0), \quad n \geq 2, \\ p_{n+1}(B_1,\ldots,B_{i-1},B_i \cup \{n+1\},B_{i+1},\ldots,B_k) = p_n(B_1,\ldots,B_k) g_n(\pi,i), \qquad i \in [k+1]. \end{aligned} \tag{1}$$

Using the natural convention $g_1(0) = 1$, the solution to these equations can be written as

$$p_n(\pi) = p_n(B_1,\ldots,B_k) = g_{\min B_2 - 1}(0) \prod_{j=\min B_2}^{n-1} g_j(\pi^{[j]}, I_j), \quad \text{where } I_j = i \text{ if } j+1 \in B_i \tag{2}$$

and $\pi^{[j]}$ is the vector of non-empty $B_i \cap [j]$. The RHS of this formula is the probability of successively creating a new first branch point when $\min B_2$ is added and inserting all higher labels such that the resulting partition at the first split is π. By the regenerative property of T_n, we can write tree probabilities as a product over branch points; for a tree $\mathbf{t} \in \mathbb{T}_n$, we identify each vertex with the set B of labels in the subtrees above this vertex, write $\pi(B)$ for the partition of the split at B, and $\widetilde{\pi}(B)$ for the partition of $[\#B]$ obtained when relabelling $\pi(B)$ by the increasing bijection from B to $[\#B]$:

$$\mathbb{P}(T_n = \mathbf{t}) = \prod_{B \in \mathbf{t}: \#B \geq 2} p_{\#B}(\widetilde{\pi}(B)) = \prod_{B \in \mathbf{t}: \#B \geq 2} \left(g_{\min \widetilde{\pi}(B)_2 - 1}(0) \prod_{j=\min \widetilde{\pi}(B)_2}^{\#B-1} g_j(\widetilde{\pi}(B)^{[j]}, I_j(B)) \right), \tag{3}$$

where $I_j(B) = i$ if $j+1 \in \widetilde{\pi}(B)_i$, and where $\widetilde{\pi}(B) = (\widetilde{\pi}(B)_1, \ldots, \widetilde{\pi}(B)_{k(B)})$.

Example 5 (Poisson-Dirichlet model [16, 18]) According to [18], the only consistent exchangeable model with splitting rules of the Gibbs form

$$p_n(\pi) = \frac{a_k}{c_n} \prod_{i=1}^{k} w_{\#B_i}, \quad \pi = (B_1,\ldots,B_k) \in \mathcal{P}_n \setminus 1_{[n]}, \qquad \text{for some } w_j \geq 0,\ a_k \geq 0,\ c_n > 0,$$

is given by a two-parameter family. Most relevant for us are $0 < \alpha < 1$ and $\theta \geq -2\alpha$ with $w_j = \Gamma(j-\alpha)/\Gamma(1-\alpha)$, $j \geq 1$, and $a_k = \alpha^{k-2}\Gamma(k+\theta/\alpha)/\Gamma(2+\theta/\alpha)$, $k \geq 2$, with normalisation constants $c_n = c_{\alpha,\theta}(n)$ satisfying $c_{\alpha,\theta}(2) = 1$ and $c_{\alpha,\theta}(n+1) = (n+\theta)c_{\alpha,\theta}(n) + \Gamma(n-\alpha)/\Gamma(1-\alpha)$, $n \geq 2$. Case $\alpha = 0$ is a limiting case. These yield growth rules for $\pi = (B_1,\ldots,B_k)$ of the form

$$\begin{aligned} g_n(0) &= p_{n+1}([n],\{n+1\}) = \frac{\Gamma(n-\alpha)}{\Gamma(1-\alpha)c_{\alpha,\theta}(n+1)}, \\ g_n(\pi,i) &= \frac{p_{n+1}(B_1,\ldots,B_{i-1},B_i \cup \{n+1\},B_{i+1},\ldots,B_k)}{p_n(B_1,\ldots,B_k)} = \frac{(\#B_i - \alpha)c_{\alpha,\theta}(n)}{c_{\alpha,\theta}(n+1)}, \quad i \in [k], \\ g_n(\pi,k+1) &= \frac{(k\alpha+\theta)c_{\alpha,\theta}(n)}{c_{\alpha,\theta}(n+1)}. \end{aligned}$$

Aldous's binary beta model [2] is included for $\theta = -2\alpha$. Both the Poisson-Dirichlet model and the alpha-gamma model contain as special cases for $\alpha \in [1/2, 1)$ and, respectively, $\theta = -1$ and $\gamma = 1 - \alpha$, the exchangeable model related to the stable Continuum Random Tree [6, 17, 19].

In the same way as the Poisson-Dirichlet model, all consistent exchangeable [15] and consistent restricted exchangeable [5] labelled Markov branching trees give rise to a regenerative tree growth rule. Section 2 develops this in terms of associated dislocation measures.

In a regenerative tree growth process $(T_n, n \geq 1)$, we can, for each $n \geq 1$, study the *residual mass process* of label 1, i.e. the evolution of the size $X_m^{(n)}$ in T_n of the block containing label 1. This process is a Markov chain in $m \geq 1$ starting from $X_0^{(n)} = n$, decreasing to $X_1^{(n)} = \#\Pi_{n,1}$ and

further according to successive splits until $M_n = \inf\{m \geq 0 \colon X_m^{(n)} = 1\}$, when label 1 becomes a singleton. We represent this Markov chain as a composition of n

$$\mathcal{C}_n = \left(C_0^{(n)}, \ldots, C_{M_n}^{(n)}\right) = \left(X_0^{(n)} - X_1^{(n)}, X_1^{(n)} - X_2^{(n)}, \ldots, X_{M_n-1}^{(n)} - X_{M_n}^{(n)}, X_{M_n}^{(n)}\right).$$

Proposition 6 *In a regenerative tree growth process, the family $(\mathcal{C}_n, n \geq 1)$ of compositions is regenerative in the sense that conditionally given $C_0^{(n)} = k$, the composition $(C_1^{(n)}, \ldots, C_{M_n}^{(n)})$ of $n-k$ has the same distribution as $\mathcal{C}_{n-k}$. The entries of the transition probability matrix are*

$$\mathbb{P}(C_0^{(n)} = n-k) = \mathbb{P}(X_1^{(n)} = k) = \sum_{\pi=(B_1,\ldots,B_k)\in\mathcal{P}_n : \#B_1 = k} p_n(\pi), \qquad 1 \leq k \leq n-1.$$

This is a straightforward consequence of Definition 1. We stress that we have consistency in the sense that $\mathcal{C}_n$ can be obtained from $\mathcal{C}_{n+1}$ by reducing one part of $\mathcal{C}_{n+1}$ by 1 (the one corresponding to label $n+1$ in T_{n+1}), but $(\mathcal{C}_n, n \geq 1)$ is not *sampling consistent* in the sense of [11] as this part is not a size-biased pick from $\mathcal{C}_{n+1}$, in general. In special cases, versions of this proposition are in the literature; in the exchangeable (sampling consistent) case, it is implicit in Bertoin's [3] study of tagged particles and explicit in [16]; for the alpha-theta model see [23, Proposition 6].

The structure of this paper is as follows. In Section 2, we encode regenerative tree growth rules $(g_n, n \geq 1)$ in a measure κ, generalising Bertoin's [3] notion of a dislocation measure. Sections 3 and 4 establish necessary and sufficient conditions on the growth rules under which results by Haas and Miermont [14, 13] apply to give scaling limits for the trees and for the residual mass processes in a regenerative tree growth process. Section 5 indicates some open problems and possible future work on regenerative tree growth.

2 Dislocation measures

Recall the notation $\mathcal{P}_n$ for the set of partitions $\pi = (B_1, \ldots, B_k)$ of $[n] = \{1, \ldots, n\}$ with blocks ordered by least element. In the Introduction, we regarded splitting rules p_n as distributions on $\mathcal{P}_n \setminus \{1_{[n]}\}$, where $1_{[n]} = ([n])$ is the partition that does not split $[n]$. In this section, we consider the set $\mathcal{P}$ of all partitions $\Gamma = (\Gamma_i, i \geq 1)$ of $\mathbb{N}$, with blocks ordered by least element and $\Gamma_i = \varnothing$ if there are fewer than i blocks. For $\Gamma \in \mathcal{P}$, we denote by $\Gamma^{[n]} \in \mathcal{P}_n$ the partition Γ restricted to $[n]$, with the non-empty $\Gamma_i \cap [n]$ as blocks. For each $\pi = (B_1, \ldots, B_k) \in \mathcal{P}_n$, $n \geq 1$, we let $\mathcal{P}^\pi = \{\Gamma \in \mathcal{P} \colon \Gamma^{[n]} = \pi\}$, the set of partitions of $\mathbb{N}$ that restrict to π on $[n]$. We equip $\mathcal{P}$ with the σ-algebra generated by $\{\mathcal{P}^\pi, \pi \in \mathcal{P}_n, n \geq 1\}$, which is also the Borel σ-algebra generated by the metric $d(\Gamma, \widehat{\Gamma}) = \exp(-\inf\{n \geq 1 \colon \Gamma^{[n]} \neq \widehat{\Gamma}^{[n]}\})$. For $\Gamma \in \mathcal{P}$ and $n \geq 1$, consider the decreasing rearrangement $|\Gamma^{[n]}|^\downarrow = (|\Gamma^{[n]}|_i^\downarrow, i \geq 1)$ of relative frequencies $(|\Gamma_i^{[n]}|, i \geq 1)$, where $|\Gamma_i^{[n]}| = \#\Gamma_i^{[n]}/n$. If the limit as $n \to \infty$ of $|\Gamma^{[n]}|_i^\downarrow$ or of $|\Gamma_i^{[n]}|$ exists, this is denoted by $|\Gamma|_i^\downarrow$ and $|\Gamma_i|$, respectively, and we say that *an asymptotic frequency exists* for that part. While the existence of some $|\Gamma_i|$ or $|\Gamma|_j^\downarrow$ does not in general imply the existence of any other $|\Gamma_k|$ or $|\Gamma|_\ell^\downarrow$, we note the following:

Lemma 7 *Existence of $|\Gamma|_i^\downarrow$ for* all *$i \geq 1$ holds if and only if $(|\Gamma_i|, i \geq 1)$ exists as a uniform limit. In this case, $(|\Gamma|_i^\downarrow, i \geq 1)$ is the decreasing rearrangement of $(|\Gamma_i|, i \geq 1)$, which we write as $|\Gamma|^\downarrow$.*

Under the equivalent conditions we say that *asymptotic ranked frequencies exist.* We leave the proof of Lemma 7 as an exercise to the reader. The partition $\Gamma = (\{2^{i-1}, \ldots, 2^i - 1\}, i \geq 1)$ is an example where $|\Gamma_i|$, $i \geq 1$, exists, but $|\Gamma|_1^\downarrow$ does not.

To study convergence to a self-similar tree $\mathcal{T}_{\gamma,\nu}$ in the sense of [12, 14], we will need to identify the scaling index $\gamma > 0$ and a measure ν on $\mathcal{S}_1^\downarrow = \{s = (s_i)_{i\geq 1} \colon s_1 \geq s_2 \geq \cdots \geq 0, \sum_{i\geq 1} s_i = 1\}$.

In the context of a regenerative tree growth process, where we have labels and consistency, these will be related to a richer structure, a dislocation measure κ on $\mathcal{P}$. This κ will provide rates

$$\lambda_n = \kappa(\{\Gamma \in \mathcal{P}\colon \Gamma^{[n]} \neq ([n])\}) = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]})$$

for the first split of $[n]$, $n \geq 2$, that allow to consistently embed the evolution of blocks in T_n, $n \geq 1$, into continuous time (see Theorem 16); the rate λ_n of the first split of $[n]$ can then be thinned by the event that this split also splits $[n-1]$, an event with probability $1-g_{n-1}(0)$, where $(g_n, n \geq 2)$ is the growth rule of the regenerative tree growth process, so that we need

$$\lambda_n(1-g_{n-1}(0)) = \lambda_{n-1}, \quad n \geq 3, \quad \text{and hence } \lambda_n = \lambda_2 \prod_{j=2}^{n-1} \frac{1}{1-g_j(0)}, \quad \text{if } g_j(0) \neq 1,\ j \geq 2. \tag{4}$$

Note that $g_j(0) = 1$ for any $j \geq 2$ means that all future insertions are made below the split concerned; if scaling limits of $(T_n, n \geq 1)$ exist at all, the parts of trees above such splits will collapse in the scaling as $n \to \infty$. We will exclude such behaviour in the sequel and make the

Assumption (A) $g_j(0) < 1$ for all $j \geq 2$.

Proposition 8 *Consider a regenerative tree growth rule $(g_n, n \geq 2)$ satisfying Assumption* **(A)** *with splitting rules $(p_n, n \geq 2)$ given by* (2)*, and let $\lambda_2 > 0$ be arbitrary. With λ_n, $n \geq 3$,* defined *by* (4)*, define*

$$\kappa(\mathcal{P}^\pi) = \lambda_n p_n(\pi), \qquad \pi \in \mathcal{P}_n \setminus \{1_{[n]}\}, n \geq 2; \qquad \kappa(\{1_{\mathbb{N}}\}) = 0. \tag{5}$$

Then κ extends uniquely to a measure on $\mathcal{P}$.

For the proof, cf. Bertoin's argument [3, Proposition 3.2] in the exchangeable case, based on Carathéodory's Extension Theorem. We refer to any measure κ on $\mathcal{P}$ with $\kappa(\{1_{\mathbb{N}}\}) = 0$ and $0 < \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}) < \infty$, $n \geq 2$, as a *dislocation measure*.

We can now condition κ on splitting $[n]$ and write p_n as

$$p_n(\pi) = \kappa(\mathcal{P}^\pi)/\kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}), \qquad \pi \in \mathcal{P}_n \setminus \{1_{[n]}\},\ n \geq 2. \tag{6}$$

We also deduce a one-to-one correspondence between κ and $((g_n, n \geq 2), \lambda_2)$:

Proposition 9 *Given a regenerative tree growth rule $(g_n, n \geq 2)$ satisfying Assumption* **(A)***, $\lambda_2 > 0$ and κ associated via* (5)*, we can recover $(g_n, n \geq 2)$ from κ via $\lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]})$, $n \geq 2$, as*

$$g_n(0) = 1 - \frac{\lambda_n}{\lambda_{n+1}}, \qquad g_n(\pi, i) = \frac{\lambda_n}{\lambda_{n+1}} \frac{\kappa(\mathcal{P}^{B_1,\ldots,B_{i-1},B_i \cup \{n+1\},B_{i+1},\ldots,B_k})}{\kappa(\mathcal{P}^{B_1,\ldots,B_k})}, \quad i \in [k+1], \tag{7}$$

$\pi = (B_1, \ldots, B_k) \in \mathcal{P}_n \setminus 1_{[n]}$. Furthermore, any dislocation measure κ yields $\lambda_2 > 0$ and a regenerative tree growth rule $(g_n, n \geq 2)$ as given in (7)*, which satisfies Assumption* **(A)***.*

We define *Kingman's paintbox* $\kappa_{\mathbf{s}}$ for $\mathbf{s} \in \mathcal{S}^\downarrow = \{\mathbf{s} = (s_i)_{i \geq 1}\colon s_1 \geq s_2 \geq \cdots \geq 0, \sum_{i \geq 1} s_i \leq 1\}$ as the distribution of the random partition Π of $\mathbb{N}$ where $i, j \in \mathbb{N}$ are in the same block if $i = j$ or $R_i = R_j \geq 1$, where the R_i, $i \in \mathbb{N}$, are independent random variables with $\mathbb{P}(R_i = k) = s_k$, $k \geq 0$, and where $s_0 = 1 - \sum_{i \geq 1} s_i$. The Strong Law of Large Numbers implies that $\kappa_{\mathbf{s}}$-a.e. $\Gamma \in \mathcal{P}$ has asymptotic ranked frequencies $|\Gamma|^\downarrow = \mathbf{s}$.

Example 10 (Exchangeable models [3, 15]) Bertoin classified all *exchangeable* dislocation measures, i.e. measures that are invariant under the action of permutations of $\mathbb{N}$ on $\mathcal{P}$, giving an integral representation

$$\kappa = \sum_{j \geq 1} c\delta_{\varepsilon(j)} + \int_{\mathcal{S}^\downarrow} \kappa_{\mathbf{s}}(\,\cdot\,)\nu(d\mathbf{s}),$$

where $c \geq 0$, $\varepsilon^{(j)}$ is the partition with blocks $\{j\}$ and $\mathbb{N}\backslash\{j\}$, and ν is a measure on $\mathcal{S}^{\downarrow}$ with

$$\nu(\{(1,0,0,\ldots)\}) = 0, \qquad \int_{\mathcal{S}^{\downarrow}} (1-s_1)\nu(d\mathbf{s}) < \infty. \tag{8}$$

Then ν is the push-forward of κ under $\Gamma \mapsto |\Gamma|^{\downarrow}$, restricted to $\mathcal{S}^{\downarrow} \setminus \{(1,0,\ldots)\}$.

The splitting rules $(p_n, n \geq 2)$ associated with Bertoin's exchangeable dislocation measures κ give rise to the consistent exchangeably labelled Markov branching trees of [15].

Example 11 (Poisson-Dirichlet model) Dislocation measures $\kappa^{\mathrm{PD}^*}_{\alpha,\theta}$ of Ewens-Pitman type are exchangeable with

$$c = 0 \quad \text{and} \quad \nu(d\mathbf{s}) = \mathrm{PD}^*_{\alpha,\theta}(d\mathbf{s}) = \mathbb{E}(\sigma_1^{\theta}; \sigma_1^{-1}\Delta\sigma_{[0,1]} \in d\mathbf{s}), \qquad \theta > -2\alpha,\ \alpha \in (0,1),$$

where $(\sigma_t, t \geq 0)$ is a stable subordinator with Laplace transform $\mathbb{E}(e^{-\lambda\sigma_t}) = e^{-t\lambda^{\alpha}}$ and $\Delta\sigma_{[0,1]}$ the decreasing rearrangement of its jumps $\Delta\sigma_t = \sigma_t - \sigma_{t-}$, $t \in [0,1]$, see [16, 18] for details. The associated $(g_n, n \geq 2)$ for the tree growth process were given in Example 5.

Example 12 (Restricted exchangeable models [5]) Let us define *restricted exchangeable* dislocation measures by their integral representation, referring to [5] for a full discussion:

$$\kappa = c_1\delta_{\varepsilon^{(1)}} + \sum_{j\geq 1}\left(c_j\delta_{\varepsilon^{(j+1)}} + k_j\delta_{\omega^{[j]}} + \int_{\mathcal{S}^{\downarrow}} \kappa_{\mathbf{s}}(\,\cdot\, \cap \mathcal{P}^{[j],\{j+1\}})\nu_j(d\mathbf{s})\right)$$

where $c_j \geq 0$, $k_j \geq 0$, $\omega^{[j]} = ([j], \{j+1\}, \{j+2\}, \ldots)$, and ν_j is a measure on $\mathcal{S}^{\downarrow}$ satisfying

$$\nu_j(\{(1,0,\ldots),(0,0,\ldots)\}) = 0, \qquad \int_{\mathcal{S}^{\downarrow}} \left(s_0 1_{\{j=1\}} + \sum_{i\geq 1} s_i^j(1-s_i)\right)\nu_j(d\mathbf{s}) < \infty, \qquad j \geq 1.$$

This includes all exchangeable dislocation measures, for $c_j = c$, $k_j = \nu(\{(0,\ldots)\})$, $\nu_j = \nu - k_j\delta_{(0,\ldots)}$. The splitting rules $(p_n, n \geq 2)$ associated with restricted exchangeable dislocation measures κ give rise to the consistent restricted exchangeable labelled Markov branching trees of [5].

Example 13 (Alpha-gamma model) The κ-measures are restricted exchangeable with

$$c_j = k_j = 0, \text{ and } \quad \nu_1(d\mathbf{s}) = (1-\alpha)\mathrm{PD}^*_{\alpha,-\alpha-\gamma}(d\mathbf{s}), \quad \nu_j(d\mathbf{s}) = \gamma\mathrm{PD}^*_{\alpha,-\alpha-\gamma}(d\mathbf{s}),\ j \geq 2,$$

$0 < \gamma < \alpha < 1$, see [5]. For the associated regenerative growth rules $(g_n, n \geq 2)$, see Example 4.

Example 14 (Alpha-theta model) This model is not restricted exchangeable and the dislocation measure is not in previous work. To describe it, we introduce an *ordered paintbox* $\widetilde{\kappa}_{(u,1-u)}$, $0 < u < 1$, as the distribution of $\Pi = (\{i \geq 1 \colon R_i = 1\}, \{i \geq 1 \colon R_i = 2\})$ where $R_1 = 1$ and the R_i, $i \geq 2$, are independent random variables with $\mathbb{P}(R_i = 1) = u = 1 - \mathbb{P}(R_i = 2)$. For $0 < \alpha < 1$ and $\theta > 0$, the κ-measure of the alpha-theta model is now given by

$$\kappa = \alpha\widetilde{\kappa}^{\mathrm{beta}}_{\theta,-\alpha}(\,\cdot\, \cap \mathcal{P}^{[2]}) + \theta\widetilde{\kappa}^{\mathrm{beta}}_{\theta,-\alpha}(\,\cdot\, \cap \mathcal{P}^{\{1\},\{2\}}), \quad \text{where } \widetilde{\kappa}^{\mathrm{beta}}_{\theta,-\alpha} = \int_0^1 \widetilde{\kappa}_{(u,1-u)}(\,\cdot\,)u^{\theta-1}(1-u)^{-\alpha-1}du.$$

To see this, let $(g_n, n \geq 2)$ be as in Example 3 and recall the convention $g_1(0) = 1$. From (2),

$$p_n(B_1, B_2) = \left(\alpha 1_{\{2\in B_1\}} + \theta 1_{\{2\in B_2\}}\right)\frac{\Gamma(\#B_1 - 1 + \theta)\Gamma(\#B_2 - \alpha)}{\Gamma(n-1+\theta)\Gamma(1-\alpha)},$$

for $(B_1, B_2) \in \mathcal{P}_n$ with $\#B_1 \geq 1$ and $\#B_2 \geq 1$, $n \geq 2$. The result now follows from (6) and the fact that $\widetilde{\kappa}^{\mathrm{beta}}_{\theta,-\alpha}(\mathcal{P}^{B_1,B_2}) = \int_0^1 u^{\theta-1+\#B_1-1}(1-u)^{-\alpha-1+\#B_2}du$ is a beta integral.

Coming back to the heuristics at the beginning of this section, let us discuss how Bertoin's [3] notion of a $\mathcal{P}$-valued homogeneous fragmentation process finds a natural extension where his exchangeable dislocation measure is replaced by a dislocation measure in the sense defined above.

Definition 15 A $\mathcal{P}$-valued process $\Pi = (\Pi(t), t \geq 0)$ is called *refining* if for all $s < t$ and all blocks $\Pi_j(t)$ of $\Pi(t)$, there is a block $\Pi_i(s)$ of $\Pi(s)$ that contains $\Pi_j(t)$. For a refining process Π, we define *genealogical trees* $T_n \in \mathbb{T}_n$, $n \geq 1$, using the representation above (3): T_n has as branch points and leaves all blocks $\Pi_i^{[n]}(t)$, $i \geq 1$, $t \geq 0$, visited by the restriction $\Pi^{[n]}$ of Π to $[n]$.

Theorem 16 *For each dislocation measure κ as defined after Proposition 8, there exists a $\mathcal{P}$-valued Feller process $\Pi = (\Pi(t), t \geq 0)$ such that the genealogical trees T_n of the restrictions $\Pi^{[n]}$ of Π to $[n]$, $n \geq 1$, form a regenerative tree growth process associated with dislocation measure κ.*

Proof. We will use κ in a Poissonian construction based on independent $\mathcal{P}$-valued Poisson point processes $(\Xi^{(i)}(t), t \geq 0)$, $i \geq 1$, with intensity measure κ. Roughly, we construct Π with $\Pi(0) = 1_{\mathbb{N}}$ such that for all i and t the partition $\Xi^{(i)}(t)$ fragments the ith block $\Pi_i(t)$ of $\Pi(t)$ into the image $\widetilde{\Xi}^{(i)}(t)$ of $\Xi^{(i)}(t)$ under the increasing bijection from $\mathbb{N}$ to $\Pi_i(t)$.

More precisely, we build consistent $\mathcal{P}_n$-valued continuous-time Markov chains $(\Pi^{[n]}(t), t \geq 0)$, $n \geq 1$, with jump times $S^{[n]}(k) \geq 0$ and jump states $M^{[n]}(k) = (M_1^{[n]}(k), \ldots, M_{K^{[n]}(k)}^{[n]}(k)) \in \mathcal{P}_n$: we set $\Pi^{[n]}(t) = M^{[n]}(k)$, $S^{[n]}(k) \leq t < S^{[n]}(k+1)$, $k \geq 0$, where $S^{[n]}(0) = 0$, $M^{[n]}(0) = 1_{[n]}$,

$$S^{[n]}(k+1) = \inf\left\{t > S^{[n]}(k) \colon [\#M_i^{[n]}(k)] \notin \Xi_1^{(i)}(t) \text{ for some } 1 \leq i \leq K^{[n]}(k)\right\}$$

and, if $S^{[n]}(k+1) < \infty$, let $M^{[n]}(k+1)$ be the partition obtained from $M^{[n]}(k)$ by replacing the ith block by the blocks of $\widetilde{\Xi}^{(i)}(S^{[n]}(k+1))$, the image of $\Xi^{(i)}(S^{[n]}(k+1)) \cap [\#M_i^{[n]}(k)]$ under the increasing bijection from $[\#M_i^{[n]}(k)]$ to $M_i^{[n]}(k)$. Note that $S^{[n]}(k+1) = \infty$ if and only if $M^{[n]}(k) = 0_{[n]} := (\{1\}, \ldots, \{n\})$, as we require $\lambda_2 = \kappa(\mathcal{P}^{\{1\},\{2\}}) > 0$ for all dislocation measures.

Since Π is uniquely determined by $(\Pi^{[n]}, n \geq 1)$, standard properties of Poisson point processes, and of the space $\mathcal{P}$ complete the proof. $\square$

By Proposition 9, growth rules $(g_n, n \geq 2)$ determine a measure κ only up to a multiplicative factor $\lambda_2 > 0$. This is reflected in the homogeneous fragmentation processes Π of Theorem 16 in the fact that the genealogical trees T_n, $n \geq 1$, are unaffected by (linear) time changes of Π.

From the consistency of $(T_n, n \geq 1)$, it is clear that there is a unique branch point of T_n, where 1 and 2 are separated into different blocks. Moreover, as n varies, the partitions at this branch point define a partition of some random subset of $\mathbb{N}$, whose distribution when relabelled by the increasing bijection is described by the splitting rules conditioned on partitions that restrict to $(\{1\}, \{2\})$, hence by $\kappa(\,\cdot\,|\,\mathcal{P}^{\{1\},\{2\}})$. In the Poissonian construction, this partition after relabelling is $\Xi^{(1)}(S^{[2]}(1))$. More generally, while there may be Poisson points that do not induce branch points of T_n, $n \geq 1$, e.g. when κ is finite or when κ can produce blocks of finite size, those points $\Xi^{(i)}(S^{[n]}(k))$ used in the Poissonian construction describe the partition at a branch point of T_m for all $m \geq n$. The partition at every branch point, separating labels j and ℓ say, has a distribution that is absolutely continuous with respect to κ.

The Poissonian construction formulated here differs from Bertoin's [3, Section 3.1.3] in the relabelling by increasing bijections: Bertoin uses $\Pi_i(t) \cap \Xi^{(i)}(t)$ instead of $\widetilde{\Xi}^{(i)}(t)$. In the exchangeable case yields the same processes, in distribution. A notable consequence is that under assumptions that ensure that there are always infinitely many blocks and that they are all infinite, we can recover the $(\Xi^{(i)}, i \geq 1)$ from Π in our setting. It is now possible to extend substantial parts of Bertoin's theory to this extended generality, but we leave the details to the reader.

3 Scaling limits for regenerative tree growth processes

Regenerative tree growth processes $(T_n, n \geq 1)$ give rise to evolutions of delabelled trees T_n°, $n \geq 1$. Denote by Q_n° the distribution of T_n° on the set $\mathbb{T}_n^\circ$ of rooted unlabelled trees with n leaves, i.e. the push-forward of the distribution of T_n under the natural projection map from $\mathbb{T}_n$

to $\mathbb{T}_n^\circ$. It is easy to verify that the family $(T_n^\circ, n \geq 1)$ has the *Markov branching property*, in the sense that conditionally given a composition $(n_1, \ldots, n_k)$, $n_1 \geq \cdots \geq n_k \geq 1$, at the first split, the distribution of T_n° is the same as if we connected independent trees $T_{n,i}^\circ$ with distribution $Q_{n_i}^\circ$, $1 \leq i \leq k$, at their roots, cf. [23, Proposition 11]. In terms of a dislocation measure κ, the probability of a composition $(n_1, \ldots, n_k)$ is given by

$$q_n(n_1, \ldots, n_k) = \sum_{\pi \in \mathcal{P}_n : (\#\pi)^\downarrow = (n_1, \ldots, n_k)} p_n(\pi) = \sum_{\pi \in \mathcal{P}_n : (\#\pi)^\downarrow = (n_1, \ldots, n_k)} \frac{\kappa(\mathcal{P}^\pi)}{\lambda_n}, \qquad \lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}).$$

Equipped with the graph metric and with the uniform probability measure ("weight measure") on the leaves, T_n° is a weighted metric space. We use notation T_n°/a to scale the metric to obtain a metric space with distance $1/a$ between any two adjacent vertices, where $a \in (0, \infty)$. Haas and Miermont [14] studied convergence of $T_n^\circ / n^\gamma \ell(n)$ for $\gamma \in (0,1)$ and ℓ slowly varying – in the sequel every ℓ will be assumed slowly varying. The sense of convergence is rooted Gromov-Hausdorff-Prohorov (GHP) convergence on the space of isometry classes of pointed weighted compact metric spaces (see e.g. [8, 9, 15, 7, 14]). The limiting random trees are rooted *γ-self-similar trees* $(\mathcal{T}, d, \rho)$ equipped with a measure μ on the Borel sets of $\mathcal{T}$, which have the property that for all $t \geq 0$ conditionally given the tree $\{v \in \mathcal{T} : d(\rho, v) \leq t\}$ up to height t and given subtree masses $\mu(S_i(t)) = m_i(t)$, the subtrees $S_i(t)$, $i \geq 1$, above height t, are like independent copies of $\mathcal{T}$, with masses rescaled by $m_i(t)$ and distances rescaled by $(m_i(t))^\gamma$.

Haas and Miermont [12] showed that the process $F(t) = (\mu(S_i(t)), i \geq 1)$, $t \geq 0$, in $\mathcal{S}^\downarrow$ is a γ-self-similar fragmentation process in the sense of Bertoin [3], and hence is associated with a dislocation measure ν on $\mathcal{S}^\downarrow$ that describes the infinitesimal behaviour of F. Specifically, a unit mass $F(0) = 1$ undergoes repeated fracturing into smaller masses that evolve independently, with a fragment of size m splitting into $m\mathbf{s} = (ms_1, ms_2, \ldots)$ at rate $m^{-\gamma}\nu(d\mathbf{s})$. It was shown in [12] that such self-similar trees exist if ν satisfies, for $\mathcal{S}_1^\downarrow = \{\mathbf{s} \in \mathcal{S}^\downarrow : \sum_{i \geq 1} s_i = 1\}$, that

$$\nu(\{(1, 0, \ldots)\}) = 0, \quad \int_{\mathcal{S}^\downarrow} (1 - s_1)\nu(d\mathbf{s}) < \infty, \quad \text{and} \quad \nu\left(\mathcal{S}^\downarrow \setminus \mathcal{S}_1^\downarrow\right) = 0. \tag{9}$$

The first two conditions, (8), are needed [3] to construct F, whereas the third condition prevents loss of mass at splitting events. The latter could in principle be captured in objects similar to Haas and Miermont's self-similar trees by allowing the measure μ to have atoms, which they disallow, but we will here follow them and exclude this case by imposing the third condition.

While every dislocation measure κ on $\mathcal{P}$ gives rise to a regenerative tree growth process, not every such process has a scaling limit. Examples without scaling limit include the $(0, \theta)$-tree growth process studied in [23, Proposition 13], where the growth is logarithmic and the branching structure degenerates under logarithmic scaling. Given a dislocation measure κ with $\lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}) = n^\gamma \ell(n)$ regularly varying, as $n \to \infty$, for some $\gamma \in (0,1)$, Haas and Miermont's [14] convergence condition **(H)** requires convergence as $n \to \infty$ of

$$\begin{aligned} &\sum_{n_1 \geq \cdots \geq n_k : n_1 + \cdots + n_k = n} n^\gamma \ell(n) q_n(n_1, \ldots, n_k) \left(1 - \frac{n_1}{n}\right) f\left(\frac{n_1}{n}, \ldots, \frac{n_k}{n}, 0, \ldots\right) \\ &= \sum_{\pi \in \mathcal{P}_n \setminus \{1_{[n]}\}} \kappa(\mathcal{P}^\pi) \left(1 - \frac{(\#\pi)_1^\downarrow}{n}\right) f\left(\frac{(\#\pi)^\downarrow}{n}\right) = \int_{\mathcal{P}} \left(1 - |\Gamma^{[n]}|_1^\downarrow\right) f\left(|\Gamma^{[n]}|^\downarrow\right) \kappa(d\Gamma), \end{aligned} \tag{10}$$

where we write $(\#\pi)^\downarrow$ for the decreasing rearrangement of the block sizes of π, with zeros appended. In view of (10), it is natural to assume the existence of asymptotic ranked frequencies κ-a.e. This holds for exchangeable and restricted exchangeable κ, and when κ is partially exchangeable in the sense of [21]. In our framework, condition **(H)** only needs checking for $f = 1$:

Theorem 17 *For a dislocation measure κ on $\mathcal{P}$ such that κ-a.e. $\Gamma \in \mathcal{P} \setminus \{1_{\mathbb{N}}\}$ has asymptotic ranked frequencies in $\mathcal{S}_1^\downarrow \setminus \{(1, 0, \ldots)\}$, define ν to be the push-forward of κ under $\Gamma \mapsto |\Gamma|^\downarrow$ and suppose $\int_{\mathcal{S}^\downarrow} (1 - s_1)\nu(d\mathbf{s}) < \infty$ and $\lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}) = n^\gamma \ell(n)$. Then the following are equivalent:*

(i) *Condition* **(H)** *holds, i.e. for all bounded continuous* $f\colon \mathcal{S}^{\downarrow} \to [0,\infty)$,

$$\int_{\mathcal{P}} \left(1 - |\Gamma^{[n]}|_1^{\downarrow}\right) f\left(|\Gamma^{[n]}|^{\downarrow}\right) \kappa(d\Gamma) \to \int_{\mathcal{S}^{\downarrow}} (1-s_1) f(\mathbf{s}) \nu(d\mathbf{s}), \qquad \text{as } n \to \infty;$$

(ii) $\displaystyle\int_{\mathcal{P}} \left(|\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right) \kappa(d\Gamma) \to 0$, *as* $n \to \infty$;

(iii) $\displaystyle\int_{\mathcal{P}} \left||\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right| \kappa(d\Gamma) \to 0$, *as* $n \to \infty$, *i.e. the convergence* $|\Gamma^{[n]}|_1^{\downarrow} \to |\Gamma|_1^{\downarrow}$ *holds in* $L_1(\kappa)$.

If condition (ii) *holds, then* $\dfrac{T_n^{\circ}}{n^{\gamma}\ell(n)} \to \mathcal{T}_{\gamma,\nu}$ *in distribution, as* $n \to \infty$, *in the GHP sense.*

Proof. If (i) holds, we obtain (ii) as a rearrangement of the special case $f = 1$. Now assume (ii). Let us prove (iii). For all $m \geq 1$, consider $\mathcal{P}^{[m]}$, all partitions that do not split $[m]$, then

$$\int_{\mathcal{P}} \left||\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right| \kappa(d\Gamma) = \int_{\mathcal{P}^{[m]}} \left||\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right| \kappa(d\Gamma) + \int_{\mathcal{P}\setminus\mathcal{P}^{[m]}} \left||\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right| \kappa(d\Gamma),$$

where for each fixed m, the second term vanishes as $n \to \infty$, by dominated convergence, since $\kappa(\mathcal{P}\setminus\mathcal{P}^{[m]}) < \infty$ and since κ-a.e. Γ has asymptotic ranked frequencies. The first term is bounded by

$$\int_{\mathcal{P}} \left(1 - |\Gamma^{[n]}|_1^{\downarrow}\right) \kappa(d\Gamma) - \int_{\mathcal{P}\setminus\mathcal{P}^{[m]}} \left(1 - |\Gamma^{[n]}|_1^{\downarrow}\right) \kappa(d\Gamma) + \int_{\mathcal{P}^{[m]}} \left(1 - |\Gamma|_1^{\downarrow}\right) \kappa(d\Gamma) \to 2 \int_{\mathcal{P}^{[m]}} (1 - |\Gamma|_1^{\downarrow}) \kappa(d\Gamma),$$

as $n \to \infty$, where the second term converges by the same reasoning as before, and the first by condition (ii). Since $\int_{\mathcal{P}} (1 - |\Gamma|_1^{\downarrow}) \kappa(d\Gamma) < \infty$, while $\bigcap_{m \geq 1} \mathcal{P}^{[m]} = \{1_{\mathbb{N}}\}$ and $\kappa(\{1_{\mathbb{N}}\}) = 0$, the infimum of these limits over $m \geq 1$ vanishes by basic measure theory, and (iii) follows.

Now assume (iii). If $f\colon \mathcal{S}^{\downarrow} \to [0,\infty)$ is continuous and bounded, then

$$\begin{aligned} &\left|\int_{\mathcal{P}} \left(1 - |\Gamma^{[n]}|_1^{\downarrow}\right) f\left(|\Gamma^{[n]}|^{\downarrow}\right) \kappa(d\Gamma) - \int_{\mathcal{P}} (1 - |\Gamma|_1^{\downarrow}) f(|\Gamma|^{\downarrow}) \kappa(d\Gamma)\right| \\ &\quad \leq \int_{\mathcal{P}} \left||\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right| \left|f(|\Gamma^{[n]}|^{\downarrow})\right| \kappa(d\Gamma) + \int_{\mathcal{P}} (1 - |\Gamma|_1^{\downarrow}) \left|f(|\Gamma^{[n]}|^{\downarrow}) - f(|\Gamma|^{\downarrow})\right| \kappa(d\Gamma), \end{aligned}$$

and (i) follows from (iii) by dominated convergence, since κ-a.e. Γ has asymptotic ranked frequencies, since f is bounded and continuous, and since ν is the pushforward of κ under $\Gamma \mapsto |\Gamma|^{\downarrow}$.

The last part follows from Haas and Miermont [14, Theorem 1]. □

Example 18 (Exchangeable models) For exchangeable $\kappa = \int_{\mathcal{S}^{\downarrow}} \kappa_{\mathbf{s}}(\cdot) \nu(d\mathbf{s})$, it was demonstrated in [14] that **(H)** reduces to the $f = 1$ case and is easily checked when (9) holds and $\lambda_n = n^{\gamma}\ell(n)$. Theorem 17 applies. This includes the Poisson-Dirichlet models of Example 5.

Example 19 (Alpha-theta model) [14] proved that their conditions hold. However, here is a shorter argument: in the notation of Example 14, we have (as [14] for exchangeable paintboxes)

$$\int_{\mathcal{P}} \left(1 - |\Gamma^{[n]}|_1^{\downarrow}\right) \widetilde{\kappa}_{(u,1-u)}(d\Gamma) \leq \int_{\mathcal{P}} \left(1 - |\Gamma_1^{[n]}|\right) \widetilde{\kappa}_{(u,1-u)}(d\Gamma) = 1 - s_1 \qquad \text{if } s_1 = u \geq 1/2,$$

and the same holds with $s_1 = 1 - u$ if $u < 1/2$. But then **(H)** for $f = 1$ follows for $\widetilde{\kappa}^{\text{beta}}_{\theta,-\alpha}$ and for κ which is bounded by a multiple of $\widetilde{\kappa}^{\text{beta}}_{\theta,-\alpha}$ by dominated convergence. Theorem 17 applies.

Example 20 (Restricted exchangeable models) Consider κ as in Example 12, with $c_j = k_j = 0$, and with $\lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}) = n^\gamma \ell(n)$. The push-forward of κ under $\Gamma \mapsto |\Gamma|^\downarrow$ is given by

$$\nu(d\mathbf{s}) = \sum_{j\geq 1} \left(\sum_{i\geq 1} s_i^j (1-s_i) \right) \nu_j(d\mathbf{s}).$$

Assuming (9), and also $\nu_j = \nu_m$ for all $j \geq m$ for some $m \geq 1$, as in [5, Theorem 7] where scaling limits were established for convergence in probability, we deduce condition (ii) of Theorem 17 for

$$\kappa = \kappa_{\nu_m} - \kappa_{\nu_m}(\,\cdot\, \cap (\mathcal{P} \setminus \mathcal{P}^{[m]})) + \sum_{j=1}^{m-1} \int_{\mathcal{S}^\downarrow} \kappa_{\mathbf{s}}(\,\cdot\, \cap \mathcal{P}^{[j],\{j+1\}}) \nu_j(d\mathbf{s}),$$

from the exchangeable case and by dominated convergence, because on the RHS only the measure $\kappa_{\nu_m} = \int_{\mathcal{S}^\downarrow} \kappa_{\mathbf{s}}(\cdot)\nu_m(d\mathbf{s})$ is infinite. Theorem 17 applies. This includes the alpha-gamma model.

All these examples satisfy the condition (ii), which we need to get convergence from Theorem 17. Let us provide another, very different example that shows what can go wrong.

Example 21 Some of the most elementary non-trivial dislocation measures are of the form

$$\kappa = \sum_{j\geq 2} (\lambda_j - \lambda_{j-1}) \delta_{\Gamma(j)} \qquad \text{for some } \Gamma(j) \in \mathcal{P}^{[j-1],\{j\}}.$$

To ensure $\lambda_n \sim n^\gamma$ for some $\gamma \in (0,1)$, let $\lambda_n - \lambda_{n-1} = \gamma n^{\gamma-1}$. For simplicity, we take $\Gamma(j)$ binary with asymptotic frequencies $(x^{(j)}, 1-x^{(j)})$, where $x^{(j)} = 1 - 1/j$. This implies

$$\int_{\mathcal{S}^\downarrow} (1-s_1)\nu(d\mathbf{s}) = \int_{\mathcal{P}} (1 - |\Gamma|_1^\downarrow)\kappa(d\Gamma) = \sum_{j\geq 2} (1 - x^{(j)})(\lambda_j - \lambda_{j-1}) = \gamma \sum_{j\geq 2} j^{\gamma-2} < \infty,$$

with ν as push-forward of κ. Consider a γ-self-similar tree $\mathcal{T}_{\gamma,\nu}$ with dislocation measure ν. We explore two examples illustrating the validity/violation of condition (ii), which now reads

$$\int_{\mathcal{P}} \left(|\Gamma^{[n]}|_1^\downarrow - |\Gamma|_1^\downarrow \right) \kappa(d\Gamma) = \sum_{j\geq 2} \left(|\Gamma(j)^{[n]}|_1^\downarrow - x^{(j)} \right) (\lambda_j - \lambda_{j-1}) \to 0, \quad \text{as } n \to \infty.$$

(a) For $j \geq 2$ and $x^{(j)} = 1 - 1/j \in (0,1)$, we construct $\Gamma(j)$ as a sequence $(\Gamma(j)^{[n]}, n \geq j)$, starting from $\Gamma(j)^{[j]} = ([j-1], \{j\})$, and using the following step inductively for $n \geq j$:

- Step A_x: Given $\Gamma^{[n]}$, if $|\Gamma_1^{[n]}| > x$, set $\Gamma_1^{[n+1]} = \Gamma_1^{[n]}$, otherwise set $\Gamma_1^{[n+1]} = \Gamma_1^{[n]} \cup \{n+1\}$.

The purpose of Step A_x is to change the relative frequency towards x. For $x = x^{(j)}$ and $\Gamma = \Gamma(j)$, we get $|\Gamma(j)_1| = |\Gamma(j)|_1^\downarrow = x^{(j)}$ and $\left| |\Gamma(j)_1^{[n]}| - x^{(j)} \right| \leq 1 - x^{(j)}$ for all $n \geq 1$, $j \geq 1$, equality for $j \geq n$ and strict inequality for $j < n$, since $1/n < 1/j = 1 - x^{(j)}$. By the Dominated Convergence Theorem, criterion (ii) of Theorem 17 is satisfied.

(b) For the convergence criterion (ii) to fail, first let $\Gamma(j)_1^{[n]}$ approach frequency $1/2$, applying Step $A_{1/2}$ for $n < a_j$, so that $|\Gamma(j)_1^{[2j]}| = 1/2$ and $|\Gamma(j)_1^{[n]}| \approx 1/2$ for $n \in [2j, a_j]$. Choose (a_j) increasing with $2 \geq \sum_{i\geq 2: n\in[2i,a_i]} (\lambda_i - \lambda_{i-1}) > 1$ and apply Step $A_{x^{(j)}}$ for $n \geq a_j$. Then we will have $|\Gamma(j)_1| = x^{(j)}$ for all $j \geq 2$, but for all n sufficiently large,

$$\sum_{j\geq 2} \left(|\Gamma(j)_1^{[n]}| - x^{(j)} \right) (\lambda_j - \lambda_{j-1}) \leq -\frac{1}{3} \sum_{j\geq 2: n\in[2j,a_j]} (\lambda_j - \lambda_{j-1}) < -\frac{1}{3} < 0.$$

Intuitively, the approximating trees have too many even branchpoints splitting into two equal-sized subtrees making trees wide and small in height, while the proposed limiting distribution produces uneven branch points leading to thin and high trees with higher probability. Gromov-Hausdorff convergence fails, if total heights do not converge [8].

4 Residual mass processes in regenerative tree growth processes

Let $(T_n, n \geq 1)$ be a regenerative tree growth process and $(X_m^{(n)}, 0 \leq m \leq M_n)$ the associated residual mass processes of label 1 in T_n, $n \geq 1$, with transition probabilities

$$\mathbb{P}(X_1^{(n)} = k) = \sum_{\pi=(B_1,\ldots,B_k)\in\mathcal{P}_n:\#B_1=k} p_n(\pi) = \frac{1}{\lambda_n}\kappa\left(\left\{\Gamma \in \mathcal{P}\colon \#\Gamma_1^{[n]} = k\right\}\right), \quad 1 \leq k < n,$$

as identified in Proposition 6, with $\lambda_n = \kappa(\{\Gamma \in \mathcal{P}\colon \Gamma_1^{[n]} \neq [n]\})$. The existence of a scaling limit for trees T_n° as studied in Section 3 does not imply the existence of a scaling limit for associated residual mass processes $X^{(n)}$, in general (see Example 24). In this section, we study scaling limits $X_{\lfloor\lambda_n t\rfloor}^{(n)}/n \to X_t$, as $n \to \infty$. Recall that positive decreasing self-similar Markov processes with $X_0 = 1$ can be represented as $X_t = \exp(-\xi_{\tau_\xi(t)})$ for a subordinator ξ and the γ-self-similar time-change $\tau_\xi(t) = \inf\{u \geq 0\colon \int_0^u \exp(-\gamma\xi_r)dr > t\} \in [0,\infty]$, $t \geq 0$. We focus on the drift-free case when $\mathbb{E}(e^{-s\xi_r}) = \exp(-r\int_{(0,\infty)}(1-e^{-sy})\Lambda(dy))$.

Given a dislocation measure κ with $\lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}) = n^\gamma \ell(n)$ regularly varying, as $n \to \infty$, for some $\gamma \in (0,1)$, Haas and Miermont's [13] condition $\mathbf{(H^*)}$ requires convergence as $n \to \infty$ of

$$\sum_{\pi=(B_1,\ldots,B_k)\in\mathcal{P}_n} n^\gamma \ell(n) p_n(\pi)\left(1-\frac{\#B_1}{n}\right) f\left(\frac{\#B_1}{n}\right) = \int_{\mathcal{P}}\left(1-|\Gamma_1^{[n]}|\right) f\left(|\Gamma_1^{[n]}|\right)\kappa(d\Gamma).$$

Theorem 22 *For a dislocation measure κ such that the first block Γ_1 of κ-a.e. $\Gamma \in \mathcal{P}$ has an asymptotic frequency $|\Gamma_1| \in (0,1)$, define Λ as push-forward of κ under $\Gamma \mapsto -\log(|\Gamma_1|)$. Suppose $\int_{(0,\infty)}(1-e^{-x})\Lambda(dx) < \infty$ and $\lambda_n = \kappa(\mathcal{P} \setminus \mathcal{P}^{[n]}) = n^\gamma \ell(n)$. Then the following are equivalent:*

(i) *Condition $\mathbf{(H^*)}$ holds, i.e. for all bounded continuous $f\colon [0,1] \to [0,\infty)$,*

$$\int_{\mathcal{P}}\left(1-|\Gamma_1^{[n]}|\right) f\left(|\Gamma_1^{[n]}|\right)\kappa(d\Gamma) \to \int_{(0,\infty)} f(e^{-y})(1-e^{-y})\Lambda(dy), \text{ as } n \to \infty;$$

(ii) $\displaystyle\int_{\mathcal{P}}\left(|\Gamma_1^{[n]}| - |\Gamma_1|\right)\kappa(d\Gamma) \to 0$, *as $n \to \infty$;*

(iii) $\displaystyle\int_{\mathcal{P}}\left||\Gamma_1^{[n]}| - |\Gamma_1|\right|\kappa(d\Gamma) \to 0$, *as $n \to \infty$, i.e. the convergence $|\Gamma_1^{[n]}| \to |\Gamma_1|$ holds in $L_1(\kappa)$.*

If condition (ii) *holds, $X_{\lfloor\lambda_n t\rfloor}^{(n)}/n \to X_t = \exp(-\xi_{\tau_\xi(t)})$ in distribution, as $n \to \infty$, in the Skorohod sense as functions of $t \geq 0$, where ξ is a subordinator with $\mathbb{E}(e^{-s\xi_r}) = \exp(-r\int_{(0,\infty)}(1-e^{-sy})\Lambda(dy))$.*

If, in addition, κ-a.e. $\Gamma \in \mathcal{P}$ has asymptotic ranked frequencies, then condition $\mathbf{(H)}$ *holds and $\dfrac{T_n^\circ}{n^\gamma \ell(n)} \to \mathcal{T}_{\gamma,\nu}$ in distribution, as $n \to \infty$, in the GHP sense.*

Proof. The proof of the equivalences is the same as for Theorem 17, with $|\Gamma^{[n]}|_1^\downarrow$ and $|\Gamma|_1^\downarrow$ replaced by $|\Gamma_1^{[n]}|$ and $|\Gamma_1|$. Convergence of $X_{\lfloor\lambda_n t\rfloor}^{(n)}/n$ is now an application of [13, Theorem 1]. Finally, if κ-a.e. $\Gamma \in \mathcal{P}$ has asymptotic ranked frequencies, we first note that in the notation of Theorem 17

$$\int_{S^\downarrow}(1-s_1)\nu(d\mathbf{s}) = \int_{\mathcal{P}}(1-|\Gamma|_1^\downarrow)\kappa(d\Gamma) \leq \int_{\mathcal{P}}(1-|\Gamma_1|)\kappa(d\Gamma) = \int_{(0,\infty)}(1-e^{-x})\Lambda(dx) < \infty. \quad (11)$$

To apply Theorem 17, we verify condition (ii) of Theorem 17:

$$\int_{\mathcal{P}}\left(|\Gamma^{[n]}|_1^\downarrow - |\Gamma|_1^\downarrow\right)\kappa(d\Gamma) = \int_{\{|\Gamma_1|=|\Gamma|_1^\downarrow\}}\left(|\Gamma_1^{[n]}| - |\Gamma_1|\right)\kappa(d\Gamma) + \int_{\{|\Gamma_1|\neq|\Gamma|_1^\downarrow\}}\left(|\Gamma^{[n]}|_1^\downarrow - |\Gamma|_1^\downarrow\right)\kappa(d\Gamma)$$
$$+ \int_{\{|\Gamma_1|=|\Gamma|_1^\downarrow\}}\left(|\Gamma^{[n]}|_1^\downarrow - |\Gamma_1^{[n]}|\right)\kappa(d\Gamma) \quad (12)$$

is a sum of three terms. The first term vanishes as $n \to \infty$ by (iii). The second term vanishes as $n \to \infty$ since $\kappa(|\Gamma_1| \neq |\Gamma|_1^{\downarrow}) < \infty$: if $|\Gamma_1| \neq |\Gamma|_1^{\downarrow}$, then one of them must be less than $1/2$, so $\kappa(|\Gamma_1| \neq |\Gamma|_1^{\downarrow}) \leq \kappa(|\Gamma_1| \leq 1/2) + \kappa(|\Gamma|_1^{\downarrow} \leq 1/2) < \infty$, by (11). The third term is non-negative, so that $\liminf_{n\to\infty} \mathrm{LHS} \geq 0$ in (12). In

$$\int_{\mathcal{P}} \left(|\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right) \kappa(d\Gamma) \leq \int_{\mathcal{P}^{[m]}} (1 - |\Gamma_1|)\, \kappa(d\Gamma) + \int_{\mathcal{P}\setminus\mathcal{P}^{[m]}} \left(|\Gamma^{[n]}|_1^{\downarrow} - |\Gamma|_1^{\downarrow}\right) \kappa(d\Gamma)$$

we can make the first term small by choosing m large and the second term vanishes as $n \to \infty$, for each fixed $m \geq 1$. Hence, $\limsup_{n\to\infty} \mathrm{LHS} \leq 0$ and so $\lim_{n\to\infty} \mathrm{LHS} = 0$. □

From the proof of Theorem 22, we can also derive a converse to the last part of Theorem 22.

Corollary 23 *In the setting of Theorem 17, the block Γ_1 containing 1 of κ-a.e. $\Gamma \in \mathcal{P} \setminus \{1_{\mathbb{N}}\}$ has an asymptotic frequency in $(0,1)$. With Λ as in Theorem 22, the following are equivalent:*

(i) $\displaystyle\int_{(0,\infty)} (1 - e^{-x})\Lambda(dx) < \infty$ *and* $\displaystyle\int_{\mathcal{P}} \left(|\Gamma_1^{[n]}| - |\Gamma_1|\right) \kappa(d\Gamma) \to 0$;

(ii) $\kappa(|\Gamma_1| \neq |\Gamma|_1^{\downarrow}) < \infty$ *and* $\displaystyle\int_{\{|\Gamma_1| = |\Gamma|_1^{\downarrow}\}} \left(|\Gamma^{[n]}|_1^{\downarrow} - |\Gamma_1^{[n]}|\right) \kappa(d\Gamma) \to 0$.

If condition (ii) *holds, $X^{(n)}_{\lfloor \lambda_n t \rfloor}/n \to X_t = \exp(-\xi_{\tau_\xi(t)})$ in distribution, as $n \to \infty$, in the Skorohod sense as functions of $t \geq 0$, where ξ is a subordinator with $\mathbb{E}(e^{-s\xi_r}) = \exp(-r \int_{(0,\infty)} (1 - e^{-sy})\Lambda(dy))$.*

It is not hard to check that these conditions are satisfied in previous examples, in which tree convergence and residual mass process convergence are known to hold, including the exchangeable case [15, Proposition 7], the particle labelled 1 in the restricted exchangeable case [5, Proposition 28] and particle 1 in the alpha-theta model [23, Proposition 6(iv)]. Let us provide an example that satisfies tree convergence in distribution and the first extra condition $\kappa(|\Gamma_1| \neq |\Gamma|_1^{\downarrow}) < \infty$, but not the second extra condition in (ii), and hence also fails the second extra condition in (i).

Example 24 In the general setting of Example 21, consider $\Gamma(j)_1^{[n]}$ that first approaches (the wrong!) frequency $1 - x^{(j)}$, applying Step $A_{1-x^{(j)}}$ for $n < a_j$, so that $|\Gamma(j)_1^{[n]}| \approx 1 - x^{(j)}$ for $n \in [j/(1 - x^{(j)}), a_j]$. Then we apply Step $A_{x^{(j)}}$ for $n \geq a_j$ to achieve $|\Gamma(j)_1| = x^{(j)}$. We call these partitions "evil". If we did this for all $j \geq 2$, too many partitions would have intermediate frequencies around $1/2$ when restricted to $[n]$ and tree convergence may fail. Note that while at $1 - x^{(j)}$, the block not containing 1 has frequency $x^{(j)}$ and is the larger block size that appears in the tree convergence criterion, while frequency $1 - x^{(j)}$ is relevant for the residual mass process.

To control the influence of partitions at intermediate frequencies, we also consider "good" partitions from Example 21(a). The following strategy gives the right mix of "good" and "evil":

1. For $j = 2$ and $j = 3$, start with two evil partitions $\Gamma(2)$ and $\Gamma(3)$, with $|\Gamma(3)_1^{[\ell]})| \approx 1 - x^{(3)}$ for $\ell = 3/(1 - x^{(3)})$, but leave a_2 and a_3 to be specified. Take good partitions $\Gamma(4), \ldots, \Gamma(\ell)$. Also recall from the general setting that $\lambda_3 - \lambda_2 = \gamma 3^{\gamma-1} > 0$. To proceed inductively, let $m = 1$, $E_1 = \{2, 3\}$, $j_1 = \ell + 1$, and proceed to step 2.

2. Given (m, E_m, j_m), release the smallest evil partition $e_m = \min E_m$ by setting $a_{e_m} = j_m$. Start evil partitions $\Gamma(j_m), \ldots, \Gamma(k_m)$ up to $k_m = \inf\{j \geq j_m : \lambda_j - \lambda_{j_m} \geq \lambda_{e_m} - \lambda_{e_m - 1}\}$. Let

$$\ell_m = \inf\left\{j \geq k_m : |\Gamma(e_m)_1^{[j]}| \approx x^{(e_m)} \text{ and } |\Gamma(i)_1^{[j]}| \approx 1 - x^{(i)}, j_m \leq i \leq k_m\right\},$$

and take good partitions $\Gamma(k_m + 1), \ldots, \Gamma(\ell_m)$. Now set $E_{m+1} = (E_m \setminus \{e_m\}) \cup \{j_m, \ldots, k_m\}$, $j_{m+1} = \ell_m + 1$ and repeat step 2. for $(m + 1, E_{m+1}, j_{m+1})$.

Now $|\Gamma(j)_1| = x^{(j)}$ for all $j \geq 2$ since $a_{e_m} < \infty$ for all evil partitions e_m. Criterion (ii) of Theorem 17 for tree convergence holds, because the good partitions and the evil partitions that are either at frequency $x^{(j)}$ or $1 - x^{(j)}$ give convergence as in Example 21(a), while the evil partitions at intermediate frequencies have total weight $w_m = (\lambda_{e_m} - \lambda_{e_m-1}) + (\lambda_{k_m} - \lambda_{j_m-1}) \to 0$ as $m \to \infty$, so their contribution vanishes as $m \to \infty$.

Criterion (ii) of Theorem 22 for residual mass process convergence is not satisfied, because for every $n \geq 3/(1 - x^{(3)})$, there are evil partitions of weight at least $\lambda_3 - \lambda_2$ which have a frequency $|\Gamma(j)_1^{[n]}| \approx 1 - x^{(j)}$ that is smaller by more than 1/4 than their limit frequency $x^{(j)}$, since $x^{(j)} - (1 - x^{(j)}) > 1/4$ for all $j \geq 3$, and this cannot be offset by partitions that exceed their limit frequencies, by the argument in Example 21(a).

5 Further problems and possible future work

Due to the coupling of $(T_n, n \geq 1)$ in a regenerative tree growth process, the convergence in distribution in Theorems 17 and 22 should be strengthened to a convergence in probability or even to almost sure convergence in all cases discussed here. We have proved tree convergence in probability in the exchangeable case [15], and in the restricted exchangeable case [5] provided that $\nu_j = \nu_m$, $j \geq m$, but the general case including the alpha-theta model remains open.

In the alpha-theta model [23] and the (restricted) exchangeable [15, 5] cases, we have established a two-stage almost sure convergence to a self-similar tree $\mathcal{T}$ by passing via reduced subtrees of T_n and of $\mathcal{T}$ spanned by the first k labelled leaves and letting first $n \to \infty$ and then $k \to \infty$. More specifically, we have embedded $(T_n, n \geq 1)$ in $\mathcal{T}$ as discrete trees with edge lengths. We have studied the evolution of reduced subtrees of $\mathcal{T}$ spanned by the first k labelled leaves, equipped with a projected mass measure, as k increases, in the alpha-theta case. We refer to the evolution of embedded trees as weighted trees with an (atomic) measure on the branches as *bead crushing* [23], see also [20] for related structures. There are several interesting variants of bead crushing processes and connections to transformations of trees. We aim to carry out a programme extending the alpha-theta case to general binary regenerative tree growth processes.

The basic embedding problem is to find a random leaf in a self-similar tree that induces a given residual mass process. Another interesting structure is the joint distribution of two residual mass processes (see [23]). When embedded in the same tree, they coincide up to a branch point and then evolve independently. We propose the terms *fragmenter* for exponential subordinators $(e^{-\xi_s}, s \geq 0)$ whose self-similar time changes are residual mass processes, and *bifurcator* for pairs of fragmenters that coincide up to an exponential time and then evolve independently.

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